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Natalis – Prenatal imaging Interpretation System

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Abstract:

Maternal mortality remains high in low-income regions ($\approx 295,000$ deaths in 2017) due to undiagnosed complications. The WHO recommends at least one ultrasound scan before 24 weeks' gestation, but up to half of pregnant women in resource-poor areas lack any screening. We propose Natalis, a smartphone-based AI ultrasound tool that analyzes only images and video to assist minimally trained health workers. Using portable ultrasound probes and deep learning, Natalis will guide the operator through a standardized scan and automatically compute gestational age, detect fetal presentation, identify placental location, count fetuses, and flag high-risk signs (e.g., placental previa, growth restriction) – similar to systems like BabyChecker and DeepEcho. Natalis is designed to run offline on an Android device with minimal training (e.g., a 3-min tutorial) and no constant power. We detail the full project plan – from data-driven model training and system architecture to hardware/software requirements and evaluation metrics (accuracy against experts, diagnostic sensitivity, etc.) – to demonstrate Natalis's potential as a deployable antenatal screening solution in low-resource settings.

Introduction:

Global Need: Approximately 295,000 women died from preventable pregnancy-related causes in 2017, with 99% of these deaths occurring in low-income countries. Major lethal complications such as hemorrhage, preeclampsia, and infections can often be anticipated by timely prenatal screening.

Ultrasound Efficacy: Ultrasound is a safe, portable, low-power imaging modality well suited to low-resource clinics. Modern handheld ultrasound probes can be battery-operated and much cheaper than CT or MRI, and integration with smartphones promises to revolutionize health by bringing ultrasound to under-resourced regions. Routine obstetric ultrasound (for example, one scan by 24 weeks) is recommended by the WHO to estimate gestational age, detect fetal anomalies, multiples, and placental issues, and improve pregnancy outcomes.

Access Gap: Despite these benefits, up to 50% of pregnant women in low-resource settings receive no ultrasound screening during pregnancy. This critical gap is largely due to a shortage of trained sonographers and reliance on outdated devices in rural clinics. Cheap portable machines are now available, but the analysis software and guidance are still lacking.

AI Solutions: Recent projects illustrate how AI can empower non-experts. For example, BabyChecker guides a health worker through a smartphone ultrasound scan and uses AI to analyze it, while DeepEcho similarly automates parts of the fetal ultrasound exam for minimally trained users. These systems demonstrate feasibility: trained and untrained operators using AI guidance can achieve expert-level measurements such as gestational age and fetal presentation from a “blind-sweep” protocol.

In summary, low-resource clinics need a self-contained ultrasound screening tool that requires only image or video input. **Natalis** will build on these advances by developing AI models and a smartphone app to make obstetric ultrasound accessible. Our solution will guide the user to acquire standard sweep scans and then analyze the images or videos to report key prenatal metrics and risk alerts.

Problem Statement:

Maternal Health Disparities and the Role of Ultrasound

A significant portion of maternal and neonatal deaths, primarily in developing regions like Sub-Saharan Africa and South Asia, stems from undetected complications during pregnancy. Conditions such as placenta previa, multiple gestation, fetal growth restriction, and preeclampsia can be diagnosed early with ultrasound, but often go unnoticed without proper screening.

The Ultrasound Access Gap

A major obstacle is the lack of resources in low-income clinics. Functional ultrasound machines are scarce and few healthcare providers are trained to use them. While the cost of portable ultrasound probes has become more affordable, the lack of ultrasound expertise remains a significant bottleneck. This resource gap means many pregnant women never receive even a single routine scan, despite the documented benefits.

Developing a Usable AI Solution

A major challenge in creating a solution is the data constraint; the models must learn solely from ultrasound images and videos, without additional sensor or manual data. The solution must also be designed with the end-user in mind: frontline health workers with minimal sonography training. Therefore, a successful tool must be simple to use, work offline, and have low power requirements. The ultimate goal is to create an AI tool that can analyze ultrasound imagery to provide critical obstetric information, such as gestational age (GA), fetal count, position, and placental location, and flag high-risk cases for referral to more advanced care.

Proposed Solution:

The **Natalis** Android app will process pre-recorded ultrasound images and videos following a standardized “blind-sweep” scanning pattern. These datasets, collected from clinical sources or open repositories, will reflect typical mid-pregnancy scans and cover the necessary fetal views for analysis. Short in-app tutorials will explain the required image or video formats and how to upload them into the system.

Image and Video Ingestion

Instead of connecting to a live ultrasound probe, Natalis will allow users to upload existing ultrasound videos or a set of still images directly from the device’s storage. All uploaded data will remain stored locally, ensuring privacy and offline operation.

On-Device AI Analysis

Using deep learning, Natalis will analyze the uploaded images and videos completely offline. The AI will perform the following key tasks:

- **Gestational Age (GA) Estimation:** Measure fetal head and abdomen to compute biometric parameters (e.g., head circumference, femur length) and estimate GA with an accuracy target of one to two weeks.
- **Fetal Presentation Detection:** Classify fetal position as cephalic (head-first) or breech (feet-first) based on anatomical orientation.
- **Placental Location Identification:** Segment the placenta to determine its position (anterior or posterior) and flag low-lying placenta cases.
- **Fetal Counting:** Detect multiple gestations by identifying more than one fetal skull or other distinguishing features.

Outputs and User Interface

After analysis, the app will display a clear summary of findings, including estimated GA, fetal presentation icons, placental location map, and fetus count. Any abnormal findings will be flagged with a clear alert (e.g., “Possible placenta previa – refer to specialist”). Representative 2D frames will be saved for patient records.

The interface will be designed for clarity and accessibility, using minimal text, intuitive icons, and support for multiple languages to ensure usability by health workers in diverse low-resource settings.

Methodology:

Proposed Natalis Development Pipeline

The Natalis project will follow a rigorous data-to-deployment pipeline to create a robust, offline-capable AI system. This approach leverages pre-existing ultrasound video and image datasets to provide critical diagnostic insights.

Data Collection and Preprocessing

The foundational step is the assembly of a high-quality **training dataset**. This will consist of ultrasound images and videos from the second trimester (14–28 weeks), when fetal anatomy is most distinct. The data will be sourced from open fetal ultrasound repositories and clinical partners, with each data point meticulously labeled with key obstetric metrics such as gestational age (GA), fetal presentation, placental location, and the number of fetuses.

To prepare this data for model training, a robust preprocessing pipeline will be implemented. Video scans will be processed to extract individual frames at standardized intervals. These frames will then be resized (e.g., to 256×256 pixels), normalized for contrast, and augmented with transformations like flipping and rotation to enhance model robustness. A filtering step will subsequently remove poor-quality frames that contain excessive noise or are not relevant to the analysis.

Advanced Model Development

A suite of specialized deep learning models will be developed using **Convolutional Neural Networks (CNNs)**, with each model addressing a specific diagnostic task. For example, a CNN classifier like **ResNet** will be trained to identify and select key standard planes (e.g., the fetal head, abdomen, and femur) within the image frames. This ensures that only the most relevant images are used for subsequent biometric measurements. A semantic segmentation network like **U-Net** will be used for fetal biometry, precisely outlining the fetal head and abdomen to compute geometric measurements for gestational age. A detection network, such as **YOLOv5**, will identify fetal skulls and bodies to determine fetal orientation (cephalic vs. breech) and count the number of fetuses. Lastly, another segmentation model will be developed to identify placental tissue, enabling accurate estimation of its position and flagging potential risks like a low-lying placenta. These models will be trained using supervised learning on GPU clusters, leveraging transfer learning from pre-trained image models to accelerate convergence and improve performance.

Block Diagram:

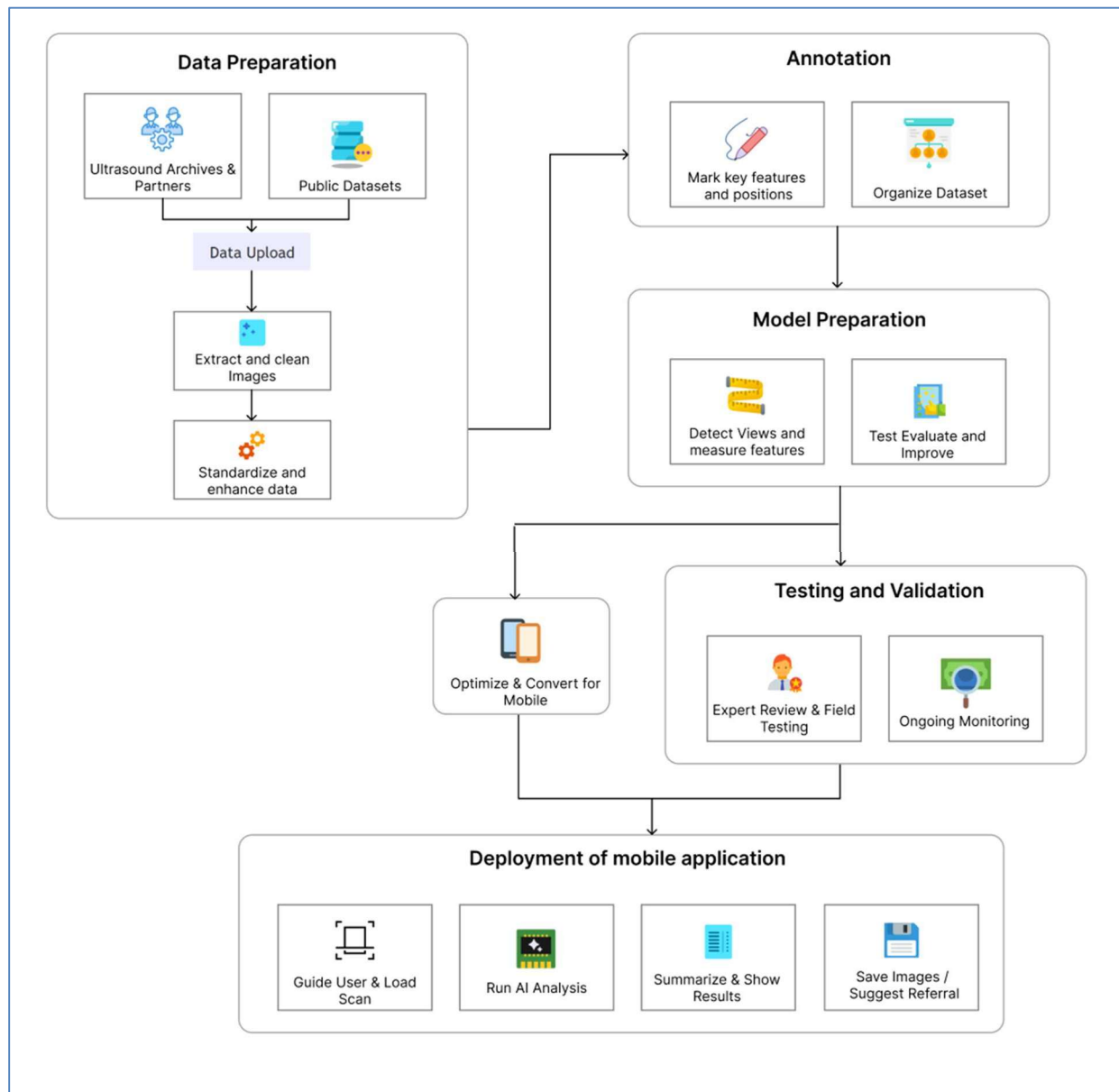


Fig. 1: Natalis High-level Block Diagram

System Diagram

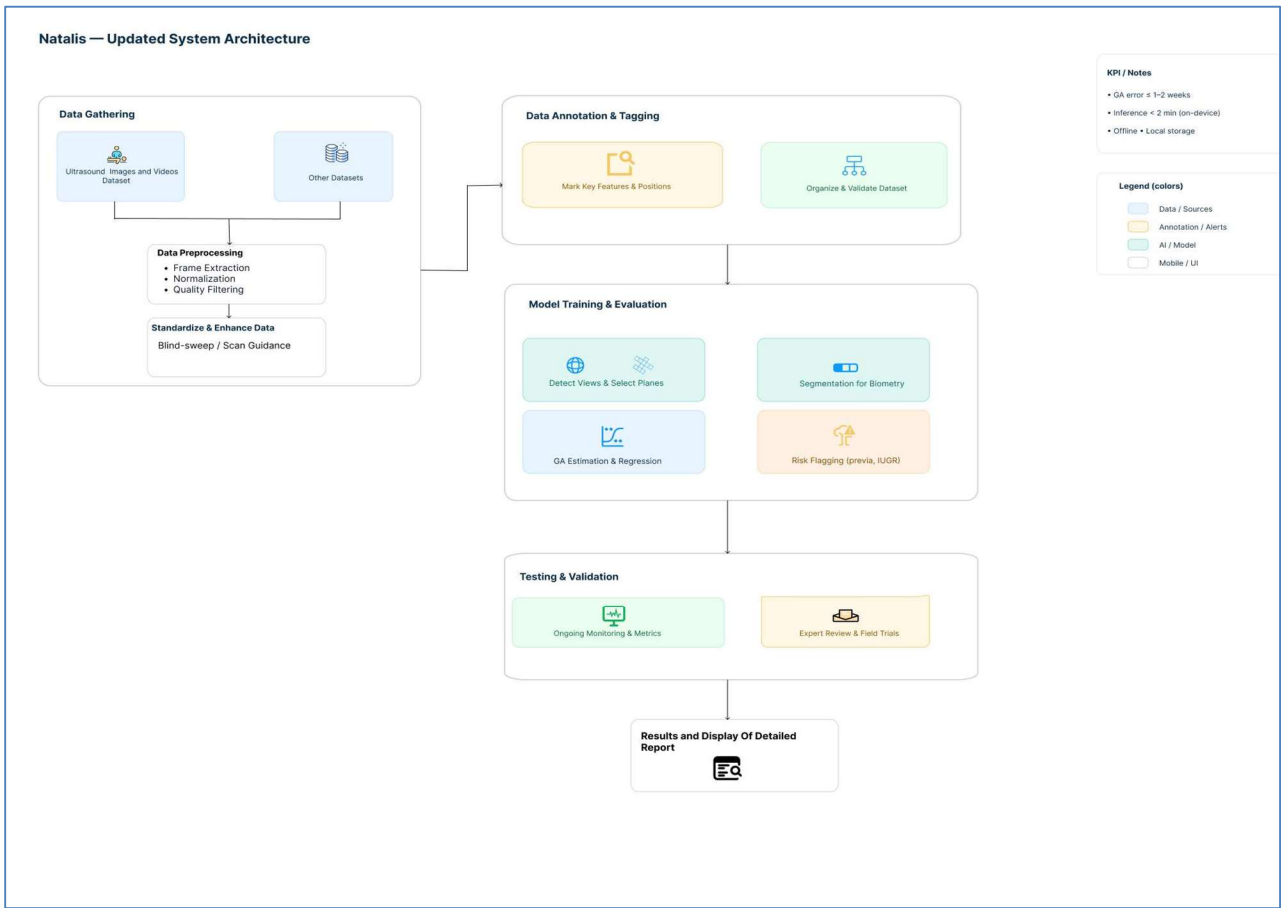


Fig. 2: Natalis System Architecture Diagram

Hardware and Software requirements

Hardware

A modern Android smartphone (Android 8+) with a multi-core CPU/GPU will run the AI models entirely offline. Long battery life or auxiliary power (e.g., power bank) is recommended. For model training, a laptop or desktop with a suitable GPU (e.g., NVIDIA RTX series) will be used.

Software & Frameworks

Model training will be done in Python using PyTorch or TensorFlow/Keras. Data processing will rely on OpenCV and NumPy. The Android app will be built in Android Studio (Java/Kotlin) and will use TensorFlow Lite or PyTorch Mobile for on-device inference. Version control will be handled with Git/GitHub, and data annotation will be done using LabelImg or a custom GUI.

Tools

We will use a dataset management system for organizing images and labels, a GPU-enabled training environment, and performance monitoring tools like TensorBoard. UI/UX prototypes will be created using mockup tools. Offline operation will be prioritized by minimizing external dependencies. CI pipelines will test model performance on each update.

By relying on readily available hardware and open-source software, Natalis will remain cost-effective and easy to reproduce in low-resource settings.

Proposed Evaluation Measures:

Quantitative Accuracy: Using a labeled test dataset, we will measure (a) Gestational Age Error – mean absolute error between the AI estimate and the obstetrician’s GA, targeting $\leq 1-2$ weeks; (b) Presentation Accuracy – classification accuracy, sensitivity, and specificity for cephalic vs. breech; (c) Placenta Localization – segmentation accuracy (e.g., Dice score) for anterior, posterior, or low-lying placenta; (d) Fetus Count – true positive rate for detecting multiple gestations.

Diagnostic Performance: We will compute precision, recall, and F1-score for risk flags (e.g., placenta previa) and benchmark toward $>95\%$ accuracy in key diagnoses.

Efficiency & Robustness: We aim for end-to-end on-device processing under 2 minutes, testing across varying image qualities and angles.

User Training: We will target a minimal training time, supported by a brief tutorial, and measure user errors or misunderstandings.

Field Simulation: In a pilot study, midwives or community health workers will use Natalis on volunteers, with results double-checked by expert sonographers. We will track referral agreement rates and gather usability feedback.

Comparison to Standards: Natalis outputs will be compared with hospital ultrasound results or birth outcomes to assess clinical reliability.

Success will be defined as achieving high agreement with expert interpretation ($>90\%$ accuracy on core tasks) while remaining practical for low-resource use.

Conclusion

Natalis aims to democratize prenatal screening by bringing AI-powered ultrasound capabilities to underserved and low-resource areas. By working entirely from existing ultrasound images and videos, and running completely offline on a smartphone, Natalis removes common barriers such as unreliable electricity, limited internet connectivity, and the shortage of highly trained sonographers.

The system's AI core will provide essential obstetric insights, including gestational age estimation, fetal position, and placental health assessment, directly at the point of care. By guiding health workers through simple steps and delivering clear, actionable results, Natalis will enable timely identification of high-risk pregnancies that require referral to advanced care.

If successfully implemented, Natalis has the potential to significantly improve antenatal care coverage in areas where access to ultrasound is currently limited. More women will be able to receive the WHO-recommended mid-pregnancy scan, and preventable complications could be detected earlier, improving outcomes for both mothers and babies.

This project plan sets out a clear pathway from data preparation and AI algorithm development to the deployment of a fully functional mobile application, laying the foundation for a cost-effective, scalable, and life-saving maternal health solution.

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